

Topic: CNN vs Visual Cortex | The Famous Cat Experiment | History of CNN

Introduction:

Before CNNs became popular in deep learning, scientists were already studying how **the human brain processes visual information**.

What they discovered about our **visual cortex** (the part of the brain responsible for vision) became the *inspiration* for how CNNs work today.

So, CNNs are not random — they are designed based on **how humans see the world** 🧐.

Human Visual Cortex

Let's start with our **visual system** (the biological side):

- The **visual cortex** is a part of the brain located at the back of your head (in the occipital lobe).
- It processes information from your eyes — turning raw visual input (light, shapes, colors) into meaningful perception (like recognizing faces, objects, etc.).

How it works:

When you see something, your **eyes capture light** → it's converted to electrical signals → sent to the **visual cortex**.

In the visual cortex:

- Some neurons respond to **edges, lines, or specific orientations** (like a vertical or horizontal line).
- Others respond to **shapes, textures, or objects**.

So, your brain processes images **hierarchically**:

1. **Lower layers** → detect simple patterns (edges, colors)
2. **Middle layers** → detect shapes or parts (like eyes, nose)
3. **Higher layers** → recognize whole objects (like a face or a car)

Does that sound familiar?

That's exactly how **CNN layers** work!

The Famous Cat Experiment (Hubel & Wiesel, 1958)

This is one of the most **legendary experiments in neuroscience**, and it directly inspired CNNs.

The Experiment:

- Conducted by **David Hubel** and **Torsten Wiesel** in 1958 at Harvard University.
 - They were studying how cats' brains respond to visual stimuli.
-

What they did:

- They showed **different shapes and patterns** (like lines, bars, and edges) to cats.
- They then **measured the activity of neurons** in the cats' visual cortex.

What they found:

- Some neurons fired **only when a line was at a specific angle** (e.g., vertical line).
- Others fired **only when a line moved** in a certain direction.
- These neurons were called **feature detectors**.

Meaning: The brain doesn't see the full image at once

It breaks it down into **simple features** (edges, lines, angles),

then **combines them** into complex patterns (shapes, faces, objects).

Conclusion from the Experiment

The key discovery:

The brain recognizes visual information **hierarchically**, layer by layer.

So:

- The **first layer** detects edges.
- The **next layer** combines edges into shapes.
- The **next** recognizes objects or faces.

This concept later became the **foundation of Convolutional Neural Networks (CNNs)**.

Each **CNN layer** does exactly what a layer of the visual cortex does:

- **Convolutional layers** → act like feature detectors
- **Pooling layers** → simplify and summarize
- **Dense layers** → make final classifications

Development of CNNs (Timeline)

Let's go through the **history of CNNs** step by step:

1958 – Hubel & Wiesel's Cat Experiment

Discovered that the brain processes images through layers of feature detectors.

Foundation of hierarchical vision processing.

1980 – Neocognitron (Kunihiko Fukushima)

- First computational model inspired by the visual cortex.
- Introduced the idea of **convolution + pooling layers**.
- Could recognize handwritten digits — but **wasn't trainable by backpropagation**.

1989 – LeNet-1 & LeNet-5 (Yann LeCun)

- First successful CNNs that used **backpropagation** for training.
- LeNet-5 was used for **handwritten digit recognition** (like postal code digits).
- Contained **convolutional, pooling, and dense layers** — just like modern CNNs!

2012 – AlexNet (Geoffrey Hinton & Alex Krizhevsky)

- Revolutionized deep learning — won the ImageNet competition.
- Used **ReLU, dropout, and GPUs** for the first time in large-scale image classification.
- Marked the **beginning of the deep learning era**.

Post-2012 – Modern CNN Architectures

- **VGGNet, GoogleNet, ResNet, DenseNet** and others built upon CNNs.
- Now used in **vision, NLP (Vision Transformers), medical AI, and robotics**.

Outro (Key Takeaways)

1. **CNNs are biologically inspired** — modeled after how our brain processes visual information.
2. **Hubel & Wiesel's cat experiment** showed that neurons detect simple → complex features in stages.
3. **This idea became the foundation** for Convolutional Neural Networks.
4. CNNs today power almost all computer vision systems.
5. The combination of **neuroscience + computer science** created one of the most powerful AI tools ever.

Quick Summary (For Fast Revision)

Concept	Explanation
Visual Cortex	Part of the brain that processes visual information
Cat Experiment (1958)	Showed that neurons detect edges, lines, and orientations
Neocognitron (1980)	Early CNN-like model (no backpropagation)
LeNet (1989)	First trainable CNN model
AlexNet (2012)	CNN that started the modern deep learning boom
CNN Inspiration	Mimics how the brain sees — from simple → complex features